INCH-POUND

MIL-G-45072G
10 November 1988
SUPERSEDING
MIL-G-45072F
24 August 1981

MILITARY SPECIFICATION

GRINDING MACHINES, DRILL, MANUAL

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers manual, floor and bench mounted twist drill point grinding machines with equipment and accessories specified herein.

1.2 Classification. The machines shall be of the following types and sizes. The type and size to be furnished shall be as specified (see 3.5 and 6.2.1).

Type I - Floor mounted

Size	Conventional drill diameter capacity range, inches
1	1/16 to 1/2
2	1/16 to 3/4
3	1/8 to 1-1/8
4	1/8 to 2
5	1/4 to 3

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Industrial Plant Equipment Center, ATTN: DIPEC-SSM, 2163 Airways Boulevard, Memphis, TN 38114-5051, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 3415

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Type II - Bench mounted

Size	Conventional drill diameter capacity range, inches
1	1/16 to 1/2
2	1/16 to 3/4
3	1/8 to 1-1/8

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issue of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2.1).

SPECIFICATIONS

MILITARY

MIL-M-18058 - Machinery, Metal and Woodworking, Packaging of.

STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Services. Federal Government.

MILITARY

MIL-STD-461 - Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.

MIL-STD-462 - Electromagnetic Interference Characteristics, Measurement of.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

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2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

CODE OF FEDERAL REGULATIONS (CFR)

U. S. DEPARTMENT OF LABOR, OCCUPATIONAL SAFETY AND HEALTH ADMINISTRATION (OSHA)

29 CFR 1910 - Occupational Safety and Health Standards (OSHA).

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402-0001.)

(Copies of specifications, standards, handbooks, drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity *or* as directed by the contracting activity.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2.1).

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI B7.1 - Safety Requirements for the Use, Care, and Protection of Abrasive Wheels. (DOD adopted)

ANSI B11.9 - Safety Requirements for the Construction, Care, and Use of Grinding Machines. (DOD adopted)

ANSI B74.2 - Specifications for Shapes and Sizes of Grinding Wheels, and Shapes, Sizes, and Identifications of Mounted Wheels. (DOD adopted)

ANSI B74.13 - Harking for Identifying Grinding Wheels and Other Bonded Abrasives. (DOD adopted)

(Application for copies should be addressed to the American National Standards Institute, ATTN: Sales Dept., 1430 Broadway, New York, NY 10018-3363.)

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NATIONAL ELECTRICAL MANUFACTURERS' ASSOCIATION (NEMA)

NEMA MG 1 - Motors and Generators. (DOD adopted)

(Application for copies should be addressed to the National Electrical Manufacturers' Association, 2101 L Street, N.W., Washington, DC 20037-1580.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 79 - Electrical Standard for Industrial Machinery. (DOD adopted)

(Application for copies should be addressed to the National Fire Protection Association, Batterymarch Park, Quincy, MA 02269-9101.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2.1), one complete grinder shall be subjected to first article inspection (see 6.3) in accordance with 4.4.

3.2 Design. The machine shall be new (not a prototype) and one of the manufacturer's current models capable of drill point grinding in accordance with the requirements herein. Type I machines shall be floor mounted and type II machines shall be bench mounted. The machine shall include all components, parts, and features necessary to meet the performance requirements specified herein. All parts subject to wear, breakage, or distortion shall be accessible for adjustment, replacement, or repair.

3.2.1 Measuring and indicating device calibrations. Measuring and indicating devices shall be graduated in inch-pound (see 6.4.1), metric (see 6.4.2), or dual (inch-pound and metric) units as specified by the procuring activity (see 6.2.1). Regardless of the measurement system used, all measuring and indicating devices on the machine shall be graduated in the same system.

3.2.1.1 Dials. All dials used to indicate machine axes position or tool movement shall be graduated in increments not larger than 0.005 inch. Dial and handwheel circumferences and scales indicating drill point and clearance angles shall be permanently and legibly engraved or etched on a nonglare background with graduation that can be read from the operator's normal position.

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3.2.2 Reclaimed materials. The machine may contain reclaimed materials to the maximum extent possible provided such materials will not jeopardize the intended use, performance, or design life of the machine. Reclaimed materials shall have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used, rebuilt, or remanufactured end products are allowed under this specification.

3.2.3 Energy efficiency. The machine and its components that directly consume energy in normal operation shall be designed and constructed for energy efficiency in accordance with the developments available within the industry.

3.2.4 Safety and health requirements. All parts, components, mechanisms, and assemblies furnished on the machine, whether or not specifically required herein, shall conform to all requirements of OSHA 29 CFR 1910 and ANSI B11.9. If a conflict arises between 29 CFR 1910 and ANSI B11.9, 29 CFR 1910 shall apply. Additional safety and health requirements shall be as specified (see 6.2.1). Covers, guards, or other safety devices normally furnished as standard on the manufacturer's commercial machine supplied to the commercial market shall be provided for the point of operation and all other parts of the machine that present safety hazards.

3.2.5 Mercury restriction. The machine shall not contain mercury or mercury compounds nor be exposed to free mercury during manufacture.

3.2.6 Asbestos restriction. Asbestos and materials containing asbestos shall not be used on or in the machine.

3.2.7 Environmental protection. During the operation, service, transportation, and storage of the machine, it shall not emit materials hazardous to the environment as prescribed by Federal, state, or local statutes in effect on the date of delivery at the point of installation.

3.2.8 Lubrication. All bearings (except for the sealed-for-life and self-lubricating types), mating gears, and sliding parts shall be provided with lubrication. Continuous lubrication shall be provided to the grinding wheel spindle. Recirculating lubrication systems shall include a filter which is cleanable or replaceable. Each lubricant reservoir shall have means for determining fluid level. All oil holes, grease fittings, and filler caps shall be accessible.

3.2.9 Interchangeability. All parts subject to replacement shall be manufactured to definite dimensions and tolerances. Replacement parts except for hand-fitted parts such as ways shall be interchangeable without requiring modification. Tooling and accessories normally supplied as standard items shall also be considered replacement parts.

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3.3 Construction. The machine shall be constructed of parts which are new, without defects, and free of repairs. The structure shall withstand all forces encountered during operation of the machine to its maximum rating and capacity without permanent distortion.

3.3.1 Castings and forgings. All castings and forgings shall be free of defects, scale, and mismatching. No processes such as weldings, peening, plugging, or filling with solder or paste shall be used for reclaiming any defective part. Such processes may be used only for enhancing surface finish and appearance.

3.3.2 Fastening devices. All fasteners shall be installed to prevent change of tightness. Fastening devices subject to removal or adjustment shall not be permanently installed.

3.3.3 Surfaces. All surfaces shall be clean and free of harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein, the condition and finish of all surfaces shall be in accordance with the manufacturer's commercial practice.

3.3.4 Melding, brazing, or soldering. Welding, brazing, or soldering shall be employed only where specified in the original design. None of these operations shall be employed as a repair measure for any defective part.

3.3.5 Painting. Unless otherwise specified (see 6.2.1), the machine shall be painted in accordance with the manufacturer's commercial practice and color.

3.3.6 Threads. All threaded parts used on the machine and its related attachments and accessories shall be US customary inch or ISO metric conforming to FED-STD-H28 and the applicable "Detailed Standard" section referenced therein.

3.3.7 Gears. All gears of spindle and axis drive trains shall be constructed from a material suitable for the intended purpose. The gears shall be of proper width and size to transmit full-rated torque and horsepower throughout the speed ranges without failure for the expected service life of the machine.

3.3.8 Electromagnetic interference control. When specified (see 6.2.1), equipment furnished under this specification shall comply with MIL-STD-461. The equipment and subsystems class and the emission and susceptibility requirements shall be as specified.

3.4 Components. The machine shall consist of not less than a base, grinding wheel assembly, drill holding device, driving mechanism, and controls. Any additional components necessary for the operation of the machine shall be furnished in accordance with the manufacturer's standard practice.

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3.4.1 Base. The base shall support the grinding wheel assembly, driving mechanism, workholding device, and controls. Type I machines shall be floor mounted machines; type II machines shall be bench mounted machines. The base on type I machines shall be a casting of iron or iron alloy or a fabricated weldment, ribbed and braced to withstand forces encountered during machine operation, with an access door or removable plate provided to service components housed within the base. Means shall be provided to secure the machine to the floor or bench, as applicable.

3.4.2 Grinding wheel assembly. The grinding wheel assembly shall include a main housing, wheel spindle and its related components, grinding wheel, and drive mechanism. The grinding wheel spindle shall be supported in radial and axial alignment by two or more sets of preloaded ball or roller bearings. The spindle shall accept a grinding wheel at one end and a driving mechanism at the other end.

3.4.2.1 Grinding wheels. Grinding wheels shall conform to ANSI B74.2. The wheels shall be marked in accordance with ANSI B74.13 and shall meet the safety requirements established in ANSI B7.1. All grinding wheels shall be properly guarded. Wheel guards shall be adjustable or detachable to allow for replacement of worn grinding wheels.

3.4.3 Workholding device. The drill holding mechanism shall be adjustable to accommodate drills of the diameters and lengths given in table I for the type and size machine specified in 1.2. A workholding chuck, collet, or a series of chucks to cover the complete range of drill sizes shall be provided. The drill holding mechanism shall oscillate the drill against the face of the grinding wheel or pass the drill across the inside diameter of a cup wheel to produce radial relief. The mechanism shall also present the drill lips at varying angles to the face of the grinding wheel to produce drill points in accordance with table I for the applicable type and size machine. A scale graduated in 1 degree increments to indicate drill point angles and a method of setting clearance angles shall be provided.

3.4.4 Drive mechanism. The grinding wheel spindle drive mechanism shall be direct, with the wheel mounted on an extension of the motor shaft or by a belt and pulley arrangement. When belt drive is supplied, the belt shall be guarded with means to adjust belt tension unless a nontensioning belt is used. Spindle drive operating speeds shall be manufacturer’s standard for the grinding wheel type and size provided with the machine.

3.4.5 Controls. All operating controls shall be located convenient to the operator’s work station(s).

3.4.6 Wheel truing device. Unless otherwise specified (see 6.2.1), each machine shall be equipped with a manual truing device capable of dressing the entire face of the grinding wheel. The device shall be complete with a commercial type diamond. The device may be an integral part of the grinding wheel assembly or independent. The device shall not interfere with drill grinding operations when not in use.

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3.4.7 Coolant system. When specified (see 6.2.1), a flood coolant system shall be provided on type I machines. The coolant system shall include a sump or reservoir, a power driven pump, and all necessary piping. The sump or reservoir shall have sufficient capacity to permit full flow of coolant. The system shall have means for draining and cleaning. Means shall be provided to permit the operator to direct and control the amount of coolant.

3.4.8 Electrical system. The electrical system on machines of less than 1-1/2 horsepower shall have a grounded plug and a double-sided on/off switch. The electrical system on machines of 1-1/2 and greater horsepower shall conform to NFPA 79 and shall have a fused safety disconnect switch or circuit breaker. An identified terminal for grounding the machine when it is installed shall be mounted in or near the disconnect switch. The terminal shall be suitable for connecting the size grounding conductor specified in NFPA 79 for" the disconnect fuse rating.

3.4.8.1 Primary input voltage. Unless otherwise specified (see 6.2.1), each machine of less than 1-1/2 horsepower shall draw all of its electrical power from a single 115-volt, 1-phase, 60-hertz circuit. Unless otherwise specified (see 6.2.1), each machine of 1-1/2 or greater horsepower shall draw all of its electrical power from a single 230/460-volt, 3-phase, 60-hertz (Hz) circuit, initially wired for operation on 230 volts.

3.4.8.2 Motors. Motors shall be rated for continuous duty and shall have ball or roller bearings of the sealed, balanced, and permanently lubricated type. Alternating-current (AC) motors shall be designed to operate on 60-Hz. Unless otherwise specified (see 6.2.1), each motor enclosure shall meet the requirements for a totally enclosed fan cooled (TEFC) motor. All motors shall meet the requirements of NEMA MG-1.

3.4.8.3 Worklight. Unless otherwise specified (see 6.2.1), a worklight shall be provided. The light shall have a protective shield, an on-off switch, and shall be adjustable to reflect maximum light on the work area. If a built-in lighting system is provided, the adjustable light described above shall not be required.

3.4.8.4 Hour meter. When specified (see 6.2.1), the machine shall be equipped with an hour meter. The hour meter shall be installed to display accumulated operating time of the main drive motor. The meter shall be of the nonresetting type and shall have a range of 0 to 99,999 hours in increments of not greater than 1 hour. The meter shall be sealed to prevent the entrance of dust and moisture and shall be mounted to withstand shock and vibration generated by the machine.

3.5 Size and capacity. Unless otherwise specified (see 6.2.1), the size and capacity of the machine shall meet the requirements of table I for the machine specified.

3.6 Performance. Each machine shall meet the drill point grinding requirements given in 4.7.3.1.

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3.7 Standard equipment. Unless otherwise specified (see 6.2.1), each machine shall be provided with all standard equipment normally furnished with the manufacturer's commercial machine.

3.8 Optional equipment. Optional equipment shall be furnished as specified and fully described (see 6.2.1).

3.9 Repair parts. Required repair parts shall be furnished as specified (see 6.2.1).

3.10 Marking on instruments, control panels, and plates. All words on instruments, control panels, and plates shall be in the English language. Characters shall be permanently marked in boldface on a contrasting background. All plates shall be corrosion-resistant.

3.10.1 Lubrication plate. When specified (see 6.2.1), a lubrication plate shall be permanently and securely attached to each machine. Unless otherwise specified (see 6.2.1), the plate shall contain the following information:

Points of lubricant application
Servicing interval
Type of lubricant
Federal or Military specification for each lubricant (if applicable)

3.10.2 Nameplate. A nameplate shall be securely attached to each machine. Unless otherwise specified (see 6.2.1), the nameplate shall contain the information listed below. If the machine is a special model, the model designation shall include the model number of the basic standard machine and a suffix identified in the manufacturer’s permanent records.

Nomenclature
Manufacturer's name
Manufacturer's model designation
Manufacturer's serial number
Power input (volts, total amps, phase, frequency)
Amp rating of motor
Short circuit/over-current protection rating
Contract Number or Order Number
National Stock Number or Plant Equipment Code
Date of manufacture

3.11 Workmanship. Workmanship of the machine shall be commensurate with all requirements specified herein and shall be of a quality equal to that prevailing among manufacturers producing equipment of the type covered by this specification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other

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facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submissions of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified (see 6.2.1), all inspections shall be performed in accordance with the test conditions specified in 4.7.1.

4.4 First article inspection. When a first article inspection is required, it shall be applied to the first article submitted in accordance with 3.1. Unless otherwise specified (see 6.2.1), first article inspection shall consist of the examination in 4.6 and all tests in 4.7 through 4.7.3.1. The machine shall pass the first article examination and all tests to be accepted.

4.5 Quality conformance inspection. Quality conformance inspection shall be applied to each item prior to being offered for acceptance under the contract. Unless otherwise specified (see 6.2.1), quality conformance inspection shall consist of the examination in 4.6, the tests in 4.7 through 4.7.3.1, and the inspection in 4.8. The machine shall pass the examination, all tests, and the inspection to be accepted.

4.6 Examination. The machine shall be examined to determine compliance with all requirements of this specification.

4.7 Tests.

4.7.1 Test conditions. All tests shall be performed in an indoor facility with an ambient temperature of 41°F to 104°F and 20 percent to 95 percent relative humidity.

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4.7.2 Operational test. The machine shall be operated at no load for not less than 30 minutes. Proper operation of all controls, motors, adjusting mechanisms, and accessories shall be verified during the trial period.

4.7.3 Performance test. Each machine shall be tested to determine its capability to grind drill points in accordance with the requirements of this specification. A two-flute high-speed twist drill of a diameter in the range for the type and size machine required shall be used for the performance test. The drill size and methods used to meet the above requirements shall be at the supplier's option. At the option of the procuring agency, each machine shall be subjected to tests in accordance with specific requirements designated by the activity in lieu of or in addition to the requirements herein (see 6.2.1).

4.7.3.1 Performance test requirements. The point angle ground shall be within ± 5 degrees of the point angle setting through the size range required in table I. Lip relief angles shall be equal within 1 degree. The point angle and lip relief angles shall be concentric with the drill body cylinder within not more than 0.002" T.I.R. The surface finish of each ground point shall be not more than 20 microinch.

4.7.4 Electromagnetic interference control test. Equipment requiring electromagnetic interference control shall be tested for compliance with 3.3.8 using the procedures given in MIL-STD-462.

4.8 Packaging inspection. Packaging of each item shall be inspected to determine compliance with the requirements of section 5.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-18058.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The grinding machines covered by this specification are intended for manually grinding drill points on fluted, straight and tapered shank twist drills used in general machine shop applications.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Type and size required (see 1.2).

c. Issue of DODISS to be cited in the solicitation and, if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).

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d. First article, if required (see 3.1).

e. If measuring and indicating devices are required to be graduated in a specific measurement system (inch-pound or metric), state required system (see 3.2.1).

f. Additional safety and health requirements, if required (see 3.2.4).

g. Painting, if different (see 3.3.5).

h. If electromagnetic interference control is required, specify the equipment and subsystem class and the emission and susceptibility required (see 3.3.8).

i. Wheel truing device, if not required (see 3.4.6).

j. Coolant system, if required (see 3.4.7).

k. Primary input voltage, if different (see 3.4.8.1).

l. Motor enclosure, if different (see 3.4.8.2).

m. Work light, if not required (see 3.4.8.3).

n. Hour meter, if required (see 3.4.8.4).

o. Size and capacity, if different (see 3.5).

p. Standard equipment, if different (see 3.7).

q. Specify and fully describe optional equipment, if required (see 3.8).

r. Specify applicable repair parts, if required (see 3.9).

s. Lubrication plate, if required; information on plate, if different (see 3.10.1).

t. Nameplate, if different (see 3.10.2).

u, Inspection conditions, if different (see 4.3).

v. First article inspection, if different (see 4.4).

w. Quality conformance inspection, if different (see 4.5).

x. Performance test, if different (see 4.7.3).

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6.2.1 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DIDs) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DIDs are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DOD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Reference Paragraph	DID Number	DID Title	Suggested Tailoring
•	•	•	•
•	•	•	•

The above DIDs were those cleared as of the date of this specification. The current issue of DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DIDs are cited on the DD From 1423.

6.2.1.1 Technical data. All technical data ordered using a DD Form 1423 should be written in the English language.

6.3 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a preproduction sample, a first article sample, a first production item, a sample selected from the first ____ production items, a standard production item from the contractor's current inventory (see 3.1), and the number of items to be tested as specified in 4.4. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 Definitions.

6.4.1 Inch-pound units. Inch-pound units are a system of measures based on the yard and pound commonly used in the United States of America and defined by the National Bureau of Standards. Inch-pound units having the same names in other countries may differ in magnitude.

6.4.2 Metric units. Metric units are a system of basic measures defined by the International System of Units based on "Le Systeme International D'Unites (SI)," of the International Bureau of Weights and Measures. These units are described in ASTM E 380 and IEEE 268 (MIL-STD-961).

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6.5 Measurement system. In this specification, all measurements, dimensions, sizes, and capacities are given in inch-pound units. These measurements may be converted to metric units through the use of the conversion factors and methods specified in FED-STD-376.

6.6 Cross-reference of classification changes. Machine classification changes from the previous issue of this specification are as follows:

MIL-G-45072F	MIL-G-45072G
Type I, Sizes 1-3	Deleted
New	Type I, Sizes 1-5
Type II, Sizes 1-4	Deleted
New	Type II, Sizes 1-3

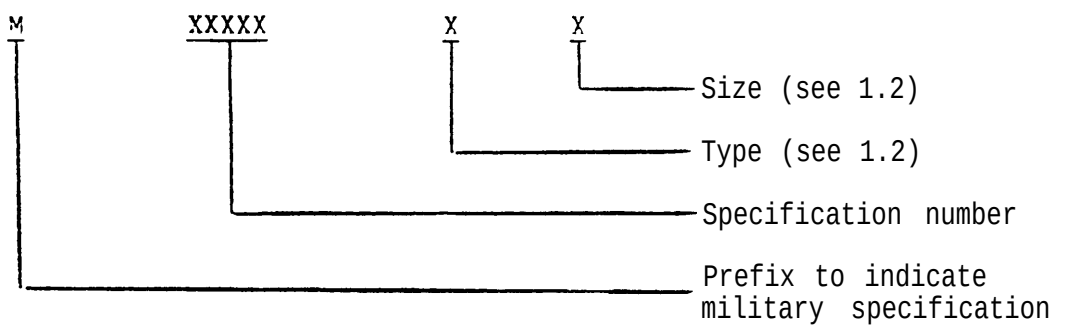
6.7 Changes from previous issues. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

6.8 Training. Training required by the procuring activity shall be provided as specified in the contract.

6.9 Warranty. Warranty requirements shall be as specified by the procuring activity in the contract.

6.10 Inspection location. The contractor should identify in his response to the solicitation the location where inspection and tests are to be performed.

6.11 Part or Identifying Number (PIN). The PIN to be used for machines acquired to this specification are created as follows:



(The Roman numerals designating machine type shall be represented in the PIN using Arabic numbers, i.e., "I" shall be represented by a "1".)

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TABLE I. Sizes and capacities.1/

Characteristic	Requirement, Type I				
	Size 1	Size 2	Size 3	Size 4	Size 5
Capacity:					
Drill diameter, range, inches	1/16 to 1/2	1/16 to 3/4	1/8 to 1-1/8	1/8 to 2	1/4 to 3
Drill length, range, inches	2 to 7	2 to 10	2 to 13	3-1/2 to 20	3-1/2 to 25
Point angle, range, degrees	90 to 180				
Lip relief angle, range, degrees	6 to 15				
Number of drill flutes	2	2	2	2, 3, or 4	
Type of drill shank	Tapered or Straight				
Motors:					
Grinding wheel, HP	1/4	1/3	1/2	1/2	1

Note 1/. Unless otherwise specified in 3.5, machine sizes and capacities shall be not less than the stated requirements of the size ordered. When a range is shown, the required performance is from the stated minimum or less to the stated maximum or greater. Machine physical size shall not be greater than the requirements of the next larger size shown in above table. When the largest size machine shown in above table is ordered, the size and capacity of the machine offered shall not exceed the stated physical size requirements by more than 10 percent.

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TABLE I. Sizes and capacities - Continued.1/

Characteristic	Requirement Type II		
	Size 1	Size 2	Size 3
Capacity:			
Drill diameter, range, inches	1/16 to 1/2	1/16 to 3/4	1/8 to 1-1/8
Drill length, range, inches	2 to 7	2 to 10	2 to 13
Point angle, range, degrees	90 to 180		
Lip relief angle, range, degrees	6 to 15		
Number of drill flutes	2		
Type of drill shank	Tapered or Straight		
Motors:			
Grinding wheel, HP	1/4	1/3	1/2

Note 1/. Unless otherwise specified in 3.5, machine sizes and capacities shall be not less than the stated requirements of the size ordered. When a range is shown, the required performance is from the stated minimum or less to the stated maximum or greater. Machine physical size shall not be greater than the requirements of the next larger size shown in above table. When the largest size machine shown in above table is ordered, the size and capacity of the machine offered shall not exceed the stated physical size requirements by more than 10 percent.

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6.12 Subject term (keyword) listing.

Bench mounted
Drill grinder
Floor mounted
Grinding machines, drill
Machine tool, grinding machines
Manual

Custodians:
Army - AL
Navy - SH
Air Force - 99

Review activities:
Air Force - 84

User activities:
Army - AV, SM
Navy - MC, YD

Preparing activity:
DLA - IP

Project (3415-0162)

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c. Reason/Rationale for Recommendation

6. REMARKS

7a. NAME OF SUBMITTER (Last, First, MI) – Optional	b. WORK TELEPHONE NUMBER (Include Area Code) – Optional
c. MAILING ADDRESS (Street, City, State, ZIP Code) – Optional	8. DATE OF SUBMISSION (YYMMDD)

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(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-G-45072C	Grinding Machines, Drill, Manual

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION (Mark one)
	☐ VENDOR
	☐ USER
b. ADDRESS (Street, City, State, ZIP Code)	☐ MANUFACTURER
	☐ OTHER (Specify):

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording

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